Chi-Squared Goodness-of-Fit Test

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Introduction

The chi-squared goodness-of-fit test compares observed counts in k categories to a pre-specified expected distribution. It answers questions like "do observed blood-type frequencies match the population reference?" or "does admission-cause distribution differ from last year's benchmark?".

Prerequisites

Categorical data, chi-squared distribution.

Theory

Given observed counts $O_1, \ldots, O_k$ and expected counts $E_j = np_j$ under H_0, where $(p_1, \ldots, p_k)$ is the specified distribution with $\sum p_j = 1$, the test statistic is

$$\chi^2 = \sum_{j=1}^{k} \frac{(O_j - E_j)^2}{E_j} \sim \chi^2_{k-1-m} \quad \text{under } H_0,$$

where m is the number of parameters estimated from the data (0 for a fully specified H_0, 1 for an estimated-mean Poisson fit, etc.).

Effect size: Cohen's $w = \sqrt{\chi^2/n}$ with thresholds 0.10, 0.30, 0.50 for small, medium, large.

Assumptions

- Independent observations.
- All expected counts ≥ 5; otherwise the chi-squared approximation breaks down and exact methods (Monte Carlo or multinomial) are preferred.
- The null distribution fully specifies the p_j (or specifies them up to estimated parameters).

R Implementation

```r
# ABO blood group audit vs. population reference
observed   <- c(A = 188, B = 42, AB = 15, O = 195)
expected_p <- c(A = 0.42, B = 0.10, AB = 0.03, O = 0.45)

chi <- chisq.test(observed, p = expected_p)
chi

chi$expected
chi$residuals

# Effect size
w <- sqrt(chi$statistic / sum(observed))
w
```

```r
# Monte Carlo p-value when expected cells are small
chisq.test(observed, p = expected_p, simulate.p.value = TRUE, B = 2000)
```

Output & Results

```
        Chi-squared test for given probabilities
X-squared = 2.18, df = 3, p-value = 0.536

       A        B       AB        O
  185.22    44.10    13.23   198.45

X-squared.
 0.065
```

No evidence the audit blood-type distribution differs from the reference (chi^2(3) = 2.18, p = 0.54, Cohen's w = 0.065, a tiny effect).

Interpretation

When the null is not rejected, do not conclude the distribution "matches"; conclude only that the data are consistent with it. Standardised residuals $(O_j - E_j)/\sqrt{E_j}$ beyond ±2 identify cells driving any rejection.

Practical Tips

- Expected counts below 5 invalidate the chi-squared approximation; use `simulate.p.value = TRUE` or an exact multinomial test.
- Report the df and effect size, not just the p-value.
- For ordered categories, consider the Cochran-Armitage trend test; chi-squared treats all categories as unordered.
- The GoF test can be used to test distributional fits (e.g., Poisson-ness) by comparing binned observed to model-expected counts; degrees of freedom decrease by the number of parameters estimated.
- Inspect standardised residuals to localise the source of any departure, rather than reporting only the omnibus p-value.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests